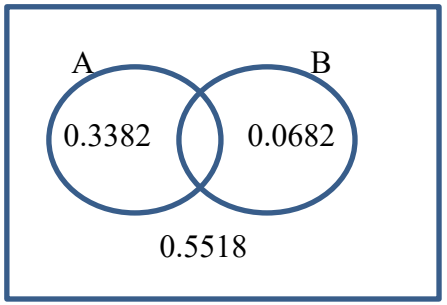
Question		Answer	Marks	AO	Guidance
7	(a)		**B1** **B1** **[2]**	**1.1** **1.1**	2 probabilities correctly placed all three correctly placed; ignore other values
7	**(b)**	[P(A ∩ B) =]$1 - 0.5518 - 0.3382 - 0.0682$ P(A) = 0.3382 + their 0.0418 **or** P(B) = 0.0682 + their 0.0418 [P(A) =]0.38 and [P(B) =] 0.11 $0.38 \times 0.11 = 0.0418$ so [A and B are] **independent**	**M1** **M1** **A1** **A1**	**3.1a** **2.1** **1.1** **3.2a**	value for P(A ∩ B) found; allow M1 for 0.0418 unsupported; allow if seen in part (a) on diagram and nowhere else may be implied by 0.38 or 0.11 even if wrongly attributed both correct; allow unsimplified allow (0.3382 + 0.0418)× (0.0682 + 0.0418) ; allow P(A)× P(B) = 0.0418 if P(A) = 0.38 and P(B) = 0.11 seen elsewhere
		Alternative method [P(A ∩ B) =]$1 - 0.5518 - 0.3382 - 0.0682$ P(B) = 0.0682 + their 0.0418 **or** P(A) = 0.3382 + their 0.0418 $P(A\|B) = \frac{0.0418}{(0.0682 + 0.0418)} = 0.38$ $P(A\|B) = 0.38 = P(A)$ so **independent**	**M1** **M1** **A1** **A1**		value for P(A ∩ B) found; allow M1 for 0.0418 unsupported allow if seen in part (a) on diagram may be implied by 0.11 **or** may be implied by 0.38 **or** $P(B\|A) = \frac{0.0418}{(0.3382 + 0.0418)} = 0.11$; allow eg $P(A\|B) = \frac{P(A \cap B)}{P(B)} = 0.38$ if P(B) = 0.11 and $P(A \cap B) = 0.0418$ seen elsewhere **or** $P(B\|A) = 0.11 = P(B)$ so independent
			[4]		

Question		Answer	Marks	AO	Guidance
7	(c)	$P(A \cap B) \neq 0$ so not mutually exclusive **or** $1 - 0.5518 \neq$ their $0.11 +$their 0.38 so $P(A \cup B) \neq P(A) + P(B)$ so not mutually exclusive	**B1** **[1]**	**2.4**	allow $P(A \cap B) =$ their 0.0418 so not mutually exclusive; must refer to probability; do **not** allow eg they have an intersection; eg the intersection is 0.0418